

Formulae

$$R = 1/CT$$

$$\text{Inventory Turns} = 1/T$$

$$T = I \times CT$$

$$I = R \times T$$

$$\text{Capacity} = \frac{m}{\text{Processing Time Per Worker}}$$

$$\text{Effective Processing Time at activity} = \frac{\text{Processing Time per worker}}{m}$$

$$\text{Break-Even Quantity} = \frac{\text{Fixed Costs}}{\text{Sales Price per Unit} - \text{Variable Cost Per Unit}}$$

$$N_{\min} = \sum t_i / CT$$

$$\text{Efficiency} = \frac{\sum t_k}{N_a \times CT}$$

$$\text{Cost of labor} = \frac{\text{Total wages per unit of time}}{\text{Flow Rate per unit of time}}$$

$$\text{Utilization} = \frac{\text{Process Flow Rate}}{\text{Capacity}} \text{ or } \frac{\text{Busy Time}}{\text{Available Time}}$$

$$\text{Utilization} = \frac{p}{m \times a}$$

$$T_q = \frac{p}{m} \frac{\text{utilization}^{\sqrt{2(m+1)}-1}}{1-\text{utilization}} \frac{CV_a^2 + CV_p^2}{2}$$

$$CV = \frac{\text{standard deviation}}{\text{mean}}$$